

RESEARCH ARTICLE

Forecast-Optimal Smoothness for Penalized Trend Estimation

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Abstract

Finite-difference penalized least squares provides a simple family of trend estimators indexed by a smoothing penalty, but the amount of smoothness is often chosen for estimation rather than for the forecast task that follows. Building on controlled-smoothness formulations of penalized trend estimation, this paper defines a normalized smoothness coordinate $S \in [0, 1]$ and selects it directly from chronological future-block forecast performance. For fixed difference order d , estimation-window length L , and forecast horizon h , the proposed criterion averages genuinely out-of-sample squared forecast errors across rolling origins and defines forecast-optimal smoothness as the minimizer of that loss. The construction preserves temporal ordering and uses the continuation implied by the same finite-difference model used to estimate the trend. We establish the monotone correspondence between the penalty parameter and normalized smoothness, derive the exact relation between smoothness and effective degrees of freedom, characterize the two limiting smoothing regimes, and show that optimization in penalty space and smoothness space is equivalent. The contribution is distinct from classical CV/GCV smoothing criteria and from kernel time-series cross-validation: the target is explicitly a horizon-specific future-block forecast loss for finite-difference penalized trends. The accompanying evaluation protocol compares this target with conventional smoothing-parameter rules under controlled trend mechanisms and public forecasting data.

KEYWORDS

forecast evaluation; penalized least squares; smoothing parameter; trend estimation; time-series cross-validation; rolling-origin forecasting

1 | INTRODUCTION

Smoothing parameters are often introduced as technical constants, although in forecasting they determine a substantive modeling choice: how much recent movement is treated as signal and how much is treated as noise. A small penalty lets the fitted trend track the observations closely, while a large penalty forces the fitted path toward a low-degree polynomial. If that trend is subsequently extrapolated, the smoothing choice also changes the future trajectory.

Finite-difference penalized least squares (PLS) makes this trade-off especially transparent. Guerrero (Guerrero, 2007, 2008) showed that the penalty can be translated into an interpretable percentage of

smoothness. This controlled-smoothness view is useful because the same numerical penalty does not represent the same effective amount of smoothing across sample sizes and difference orders. The remaining practical question is immediate: if the fitted trend will be used for forecasting, which percentage of smoothness should be chosen?

The forecasting literature provides strong precedents for choosing tuning parameters by predictive performance. Forecast-error minimization has long been used to estimate exponential-smoothing parameters (Taylor, 2004); time-series cross-validation selects a kernel bandwidth through one-step-ahead prediction in dependent data (Hart, 1994); and recent work in the *Journal of Forecasting* continues to connect hyperparameter selection, trend filtering, and out-of-sample

loss to forecast performance (Zafar, Kellard, & Vinogradov, 2022; Staněk, 2023; Wolff & Echterling, 2024; Franjic & Schweikert, 2025; Xu, Aschakulporn, & Zhang, 2025). These papers motivate forecast-oriented tuning, but they do not define the smoothness percentage of a finite-difference penalized trend through a horizon-specific rolling future-block loss.

For fixed difference order d , estimation-window length L , and forecast horizon h , let $S \in [0, 1]$ denote normalized smoothness. At each historical forecast origin, the trend is fitted using only observations available at that origin, extrapolated by the continuation implied by the same finite-difference model, and scored against the next h realized observations. Averaging these errors yields a chronological criterion $F_{d,L,h}(S)$. We define

$$S_{d,L,h}^* \in \arg \min_{S \in [0,1]} F_{d,L,h}(S) \quad (1)$$

as the *forecast-optimal smoothness* for that forecasting configuration.

The proposal is deliberately narrower than a general automatic forecasting system. The order d , window length L , and horizon h are treated as fixed when defining the continuous smoothness problem. Joint adaptation of all three quantities is a different model-selection problem. Likewise, the present paper defines and studies the forecast target; a companion numerical study treats the separate problem of efficiently recovering multiple minima when the one-dimensional objective is non-unimodal.

The distinction from existing smoothing criteria is also important. Classical cross-validation and generalized cross-validation are usually motivated by estimation or reconstruction performance, while information criteria optimize different statistical objectives. Cortés-Toto et al. (Cortés-Toto, Guerrero, & Reyes, 2017) study the smoothness achieved by several such selectors inside PLS. The criterion proposed here is not intended to dominate those methods universally. It answers a different question: conditional on using the penalized trend as a forecasting device with a specified continuation and horizon, which amount of smoothness minimizes chronological forecast loss?

The contribution has four parts. First, controlled smoothness is converted into an endogenous forecast-based selection rule. Second, the normalized $S \in [0, 1]$ coordinate makes the unsmoothed and maximally smoothed finite-difference limits explicit. Third, the target is horizon-specific through genuine future blocks rather than only one-step reconstruction or leave-one-out error. Fourth, tuning is embedded in a strictly

chronological rolling-origin protocol, so the information set matches actual forecast use.

The remainder of the paper positions the proposal relative to controlled smoothness, predictive smoothing selection, and forecast evaluation; develops the finite-difference PLS representation; defines the forecast-optimal criterion; establishes its basic properties; and specifies the simulation and empirical protocol required to evaluate when forecast-based smoothness differs materially from conventional alternatives.

2 | RELATED WORK AND POSITIONING

2.1 | Controlled smoothness in finite-difference PLS

The direct methodological foundation is the controlled-smoothness formulation of finite-difference PLS. Guerrero (Guerrero, 2007) derives a statistical representation of the penalized trend estimator and introduces a trace-based smoothness index linking the penalty parameter to an interpretable amount of smoothness. Guerrero (Guerrero, 2008) develops this into a user-controlled trend-estimation procedure in which a desired smoothness level is chosen first and the corresponding penalty is recovered afterward.

Cortés-Toto et al. (Cortés-Toto et al., 2017) study what smoothness is produced when the PLS penalty is selected by CV, GCV, AICc, and BIC. Their results make an important distinction for the present paper: a rule for selecting λ and the amount of smoothness ultimately induced by that rule are not the same object.

The present paper reverses the controlled-smoothness direction:

$$\text{user chooses } S \longrightarrow \lambda(S)$$

becomes

$$\text{chronological forecast loss} \longrightarrow S^*.$$

The percentage of smoothness remains the interpretable object, but it is selected endogenously for the forecast task.

2.2 | Predictive selection of smoothing and tuning parameters

Forecast-based tuning is established in other smoother families. Hart (Hart, 1994) considers kernel smoothing with serially correlated errors and proposes time-series cross-validation (TSCV), selecting the bandwidth through one-step-ahead predictive performance. This is an important conceptual precedent because it distinguishes good trend recovery from good prediction under dependence.

The object considered here is different. Hart selects a kernel bandwidth and explicitly models serial dependence in the errors. We select the smoothness of a finite-difference PLS trend, retain the model's native continuation rule, and allow an arbitrary h -step future block to define the tuning target. The connection to Hart is therefore the predictive principle rather than estimator equivalence.

Forecast-error tuning is also standard in exponential smoothing. Taylor (Taylor, 2004) describes data-driven smoothing-parameter estimation through ex post one-step-ahead forecast errors and develops an adaptive smooth-transition formulation. More recent *Journal of Forecasting* papers apply the same predictive principle to other tuning problems. Wolff and Echterling (Wolff & Echterling, 2024) choose regularization and machine-learning tuning parameters using validation forecast errors. Franjic and Schweikert (Franjic & Schweikert, 2025) propose a cross-validation strategy that connects predictor preselection directly to nowcast performance. Xu et al. (Xu et al., 2025) compare bandwidth-selection schemes in a time-varying-parameter HAR model and show that bandwidth choice affects forecast performance.

These papers establish that forecast-oriented tuning is a legitimate methodological target. The narrower gap addressed here is the selection of a controlled smoothness percentage inside the finite-difference PLS family.

2.3 | Trend filtering for forecasting

Zafar et al. (Zafar et al., 2022) explicitly evaluate trend-filtering methods by the short-term forecasts they enable. Their study reinforces the idea that a trend representation should be judged partly by predictive consequences rather than only by visual smoothness or in-sample fit. Our objective is complementary: rather than introduce a new trend representation, we keep finite-difference PLS fixed and optimize the amount of smoothness within that family.

2.4 | Chronological forecast evaluation

The proposed criterion uses pseudo-out-of-sample loss as a tuning objective. Tashman (Tashman, 2000) reviews out-of-sample forecast evaluation schemes, while Bergmeir and Benítez (Bergmeir & Benítez, 2012) discuss cross-validation for time-series predictors. Staněk (Staněk, 2023) formalizes out-of-sample loss estimation and distinguishes rolling and fixed evaluation schemes. His formulation also makes explicit that the relevant forecast loss depends on the forecast horizon, model memory, and updating design.

This is closely aligned with $F_{d,L,h}(S)$. Smoothness is selected for a specified horizon and update protocol, and the information set available at each origin is part of the estimator's definition.

3 | FINITE-DIFFERENCE PENALIZED TREND ESTIMATION

3.1 | Penalized least squares

At forecast origin T , let

$$x_T = (y_{T-L+1}, \dots, y_T)^\top \in \mathbb{R}^L$$

denote the estimation window. For difference order $d < L$, let

$$D_d \in \mathbb{R}^{(L-d) \times L}, \quad Q = D_d^\top D_d.$$

The baseline estimator solves

$$\hat{\tau}_T(\lambda) = \arg \min_{\tau \in \mathbb{R}^L} \{ \|x_T - \tau\|_2^2 + \lambda \|D_d \tau\|_2^2 \}, \quad \lambda \geq 0, \quad (2)$$

with solution

$$\hat{\tau}_T(\lambda) = H_\lambda x_T, \quad H_\lambda = (I + \lambda Q)^{-1}. \quad (3)$$

The paper uses the zero-drift, identity-weight formulation in (2) to isolate the smoothness-selection question. Guerrero's broader statistical formulation allows nonzero mean differences and more general covariance weighting (Guerrero, 2007); these are natural extensions but are not silently imposed in the baseline criterion.

3.2 | Spectral representation

Because

$$v^\top Q v = \|D_d v\|_2^2 \geq 0,$$

Q is symmetric positive semidefinite. Under the standard difference operator,

$$\text{rank}(Q) = L - d,$$

so Q has d zero eigenvalues and $L - d$ positive eigenvalues. Write

$$Q = U \text{diag} \left(\underbrace{0, \dots, 0}_d, \delta_1, \dots, \delta_{L-d} \right) U^\top, \quad \delta_j > 0.$$

Then

$$H_\lambda = U \text{diag} \left(\underbrace{1, \dots, 1}_d, \frac{1}{1 + \lambda \delta_1}, \dots, \frac{1}{1 + \lambda \delta_{L-d}} \right) U^\top. \quad (4)$$

Thus the d null-space directions are reproduced without shrinkage, while each penalizable direction is attenuated by

$$\alpha_j(\lambda) = \frac{1}{1 + \lambda \delta_j}.$$

For $d = 2$, the null space consists of discrete linear sequences, so increasing λ moves the fitted path toward a linear trend.

3.3 | Normalized smoothness

Guerrero's trace-based construction motivates measuring smoothing through the effective action of H_λ . We use

$$S(\lambda) = 1 - \frac{1}{L-d} \sum_{j=1}^{L-d} \frac{1}{1 + \lambda \delta_j}. \quad (5)$$

The normalization excludes the d directions that cannot be penalized, so

$$S(0) = 0, \quad \lim_{\lambda \rightarrow \infty} S(\lambda) = 1.$$

For a fixed linear smoother,

$$\text{edf}(\lambda) = \text{tr}(H_\lambda).$$

Combining (4) and (5),

$$\boxed{\text{edf}(\lambda) = L - (L - d)S(\lambda)}. \quad (6)$$

Hence S is a normalized complement of effective degrees of freedom.

4 | FORECAST-OPTIMAL SMOOTHNESS

4.1 | Continuation of the fitted trend

A trend estimate does not become a forecast until a continuation rule is specified. In the baseline model, the fitted finite-difference structure is continued by imposing zero future d th differences. This yields a fixed linear operator

$$G_{d,h} \in \mathbb{R}^{h \times L}$$

such that

$$\hat{z}_T(\lambda) = G_{d,h} H_\lambda x_T. \quad (7)$$

Equivalently, using backward differences at the end of the fitted trend,

$$\hat{\tau}_{T+k|T} = \sum_{j=0}^{d-1} \binom{k+j-1}{j} \nabla^j \hat{\tau}_T, \quad k = 1, \dots, h. \quad (8)$$

For $d = 1$ the continuation is constant; for $d = 2$ it is linear; and for $d = 3$ it is quadratic. This continuation is the extrapolation implied by the same finite-difference structure that defines the fitted trend.

4.2 | Rolling future-block loss

Let

$$z_T = (y_{T+1}, \dots, y_{T+h})^\top$$

be the realized future block at origin T . For candidate λ ,

$$f_T(\lambda) = \frac{1}{h} \|z_T - G_{d,h} H_\lambda x_T\|_2^2. \quad (9)$$

For chronological origins $\mathcal{O} = \{T_1, \dots, T_M\}$, define

$$f_{d,L,h}(\lambda) = \frac{1}{M} \sum_{m=1}^M f_{T_m}(\lambda). \quad (10)$$

No observation after T_m may affect the fit or forecast at that origin; future observations are revealed only for scoring.

Because $S(\lambda)$ is one-to-one, define

$$F_{d,L,h}(S) = f_{d,L,h}(\lambda(S)), \quad S \in [0, 1]. \quad (11)$$

The proposed selector is

$$\boxed{S_{d,L,h}^* \in \arg \min_{S \in [0,1]} F_{d,L,h}(S)}. \quad (12)$$

We call (12) *forecast-optimal smoothness*. Optimality is conditional on the specified PLS family, difference order, window length, continuation rule, horizon, rolling-origin design, and loss.

4.3 | Why future blocks rather than only one-step errors

A one-step criterion is appropriate when the operational task is one-step forecasting. For longer horizons, however, the smoothness choice affects both terminal derivatives and how those derivatives accumulate under extrapolation. The relevant tuning target can therefore change with h .

The future-block construction makes this dependence explicit. If the decision problem evaluates h future observations jointly, smoothness is selected against that same block rather than against a reconstruction surrogate.

4.4 | Relationship to conventional cross-validation

The criterion in (11) evaluates observations excluded from the fit at each origin, but it is neither ordinary leave-one-out CV nor GCV. Classical smoothing CV/GCV usually evaluates reconstruction under the same sample geometry; the proposed criterion evaluates extrapolation into a genuinely later block.

This also separates the proposal from Hart's TSCV (Hart, 1994). Both use predictive performance to tune smoothness, but Hart's object is a kernel bandwidth under one-step prediction with a time-series error model. Here the object is normalized finite-difference smoothness and the forecast is the native continuation of the penalized trend over an arbitrary horizon.

5 | BASIC PROPERTIES

Proposition 1 (Monotone smoothness reparameterization). *For positive penalized eigenvalues $\delta_1, \dots, \delta_{L-d}$,*

$$S'(\lambda) = \frac{1}{L-d} \sum_{j=1}^{L-d} \frac{\delta_j}{(1 + \lambda\delta_j)^2} > 0,$$

and

$$S''(\lambda) = -\frac{2}{L-d} \sum_{j=1}^{L-d} \frac{\delta_j^2}{(1 + \lambda\delta_j)^3} < 0.$$

After adjoining $\lambda = \infty$, S maps $[0, \infty]$ continuously and strictly increasingly onto $[0, 1]$.

Proof. Differentiate (5) term by term. Positivity of each δ_j gives the signs; the endpoint values follow from $(1 + \lambda\delta_j)^{-1}$. $\square$

The bounded scale has a common interpretation across fixed (d, L) configurations while preserving the ordering of penalty strength.

Proposition 2 (Effective degrees of freedom). *For fixed λ ,*

$$\text{tr}(H_\lambda) = L - (L - d)S(\lambda).$$

Proof. From (4),

$$\text{tr}(H_\lambda) = d + \sum_{j=1}^{L-d} (1 + \lambda\delta_j)^{-1}.$$

Substitution of (5) yields the result. $\square$

Proposition 3 (Equivalent optimization in penalty and smoothness space). *Let $F(S) = f(\lambda(S))$. Then the global minimizers in $\lambda \in [0, \infty]$ and $S \in [0, 1]$ correspond under the bijection in Proposition 1. At interior points,*

$$F'(S) = \frac{f'(\lambda)}{S'(\lambda)},$$

so interior stationary points correspond exactly.

Proof. A strictly increasing bijection preserves the minimizing set. The derivative identity follows from the chain rule. $\square$

Proposition 4 (Endpoint interpretation). *At $S = 0$, $H_\lambda = I$ and the historical trend reproduces the observations. At $S = 1$, interpreted as $\lambda \rightarrow \infty$, H_λ converges to the orthogonal projection onto $\ker(D_d)$.*

For $d = 2$, the endpoints are therefore the unsmoothed historical path and its least-squares linear projection. Endpoint comparison is part of the scientific selection problem, not merely a numerical detail.

5.1 | Existence and possible nonuniqueness

For finite $S < 1$, $H_{\lambda(S)}$ is continuous. The limiting matrix at $S = 1$ exists by (4). Thus the forecast loss has a continuous extension to the compact domain $[0, 1]$ and at least one global minimizer exists.

Uniqueness is not assumed. Distinct smoothing regimes can produce similar validation loss, and the criterion may contain several local minima. This is statistically meaningful because competing minima can imply different effective degrees of freedom and terminal slopes. Efficient recovery of all numerical candidates is delegated to the companion numerical-methods paper; the definition in (12) does not depend on a particular optimizer.

5.2 | Forecast optimality versus recovery optimality

If a latent trend τ is known in simulation, a recovery-oriented oracle might choose

$$S_{\text{rec}} \in \arg \min_S \|\hat{\tau}_S - \tau\|^2.$$

The proposed criterion instead chooses $S_{d,L,h}^*$ from future predictive error. These targets need not coincide. A smoother can reconstruct the historical latent path accurately but extrapolate poorly, or sacrifice historical recovery in exchange for more stable terminal derivatives.

This distinction is a central empirical question of the paper, not a claim that forecast-optimal smoothness is universally preferable under every loss.

6 | EVALUATION PROTOCOL

The final submission should evaluate the criterion as a forecasting method, not only as a mathematical reparameterization. This section fixes the comparison logic before final test results are inserted.

6.1 | Simulation questions

The simulation study is organized around three questions:

1. Does forecast-CV select a different amount of smoothness from recovery-oriented or conventional in-sample selectors?
2. When the selected smoothness differs, does the horizon-matched choice improve genuinely held-out forecast loss?
3. How do sample size, noise dependence, trend roughness, and forecast horizon change the selected S ?

The baseline simulations use $d = 2$ and fixed estimation windows, with $d = 1$ and $d = 3$ reserved for robustness. Each generated series is split chronologically into a smoothness-selection segment and a later test segment. All tuning occurs before the test observations are revealed.

The latent trend mechanisms should include a nearly linear trend; a smoothly changing slope; a piecewise-smooth trend with a level or slope change; and a low-frequency nonlinear trend without an abrupt break.

Observation noise is varied across independent Gaussian noise, persistent AR(1) noise, and a heavy-tailed robustness condition. Sample size and signal-to-noise ratio are crossed with the trend mechanisms.

For each design, forecast horizons are prespecified rather than selected from the same test data. A central comparison is whether $S_{d,L,h}^*$ changes materially as h changes.

6.2 | Competing smoothness selectors

The proposed horizon-matched rolling criterion is compared with established PLS selectors:

1. future-block forecast-CV defined in (11);
2. ordinary CV;
3. GCV;
4. AICc;
5. BIC;
6. selected fixed percentages of controlled smoothness used only as interpretable reference points.

The CV/GCV/AICc/BIC set follows the PLS comparison studied by Cortés-Toto et al. (Cortés-Toto et al., 2017). Their study asks how much smoothness these criteria induce; ours asks how the selected trends subsequently forecast when extrapolated.

A second comparison isolates horizon matching. For $h > 1$, the same trend family is tuned once using one-step rolling loss and once using the h -step future-block loss. This directly tests whether matching the tuning horizon to the operational horizon matters.

6.3 | Evaluation metrics

For simulations with known latent trend, report both

trend-recovery MSE and future forecast MSE.

This prevents a forecast-oriented selector from being judged only by its own objective and makes the recovery-versus-prediction distinction observable.

For real data, the primary endpoint is held-out forecast MSFE. Secondary summaries include MAE, selected smoothness S , effective degrees of freedom from (6), and the stability of selected smoothness across adjacent forecast origins.

When comparing many series, report both per-series relative loss and aggregate summaries. Formal predictive-ability tests should be used only when their dependence assumptions match the evaluation design.

6.4 | Public real-data evaluation

The submission-stage empirical analysis should emphasize reproducibility and chronological integrity. Two complementary panels are suitable.

First, a broad public forecasting panel such as the M4 data can evaluate whether the criterion behaves sensibly across heterogeneous series, following the forecast-evaluation tradition represented by Staněk (Staněk, 2023). Series that require seasonal preprocessing should use a prespecified transformation so that the paper remains about trend smoothness rather than seasonal-model selection.

Second, a small macroeconomic panel can provide interpretable trend examples. Historical macroeconomic backtests require real-time vintages; revised current histories are acceptable only for exploratory analysis because they can contain information unavailable at the historical forecast origin.

The repository already distinguishes exploratory external-validation snapshots from paper-final vintage requirements. That distinction must be preserved in the reported results.

6.5 | Result-freeze rule

All comparison rules, forecast origins, horizon sets, and primary metrics should be fixed before inspecting the final test block. Numerical optimization settings must not be tuned from test performance. The companion numerical-methods paper may provide a more efficient optimizer, but the scientific definition of the criterion is unchanged if a sufficiently fine deterministic grid is used for the present paper.

7 | SCOPE, INTERPRETATION, AND IMPLICATIONS

7.1 | What the selected smoothness means

The output $S_{d,L,h}^*$ is a forecast-task-specific complexity choice. From (6),

$$S^* = \frac{L - \text{edf}^*}{L - d},$$

so the selected value has a direct effective-degrees-of-freedom interpretation.

A value near zero says that, within the specified finite-difference family and validation design, preserving most observed local variation improved the forecast criterion. A value near one says that the criterion preferred a trend close to the null-space polynomial. Intermediate values represent partial attenuation of penalizable spectral directions.

This interpretation is more comparable across sample lengths than reporting λ alone because the same numerical penalty can imply different attenuation patterns when L changes.

7.2 | What forecast-optimal smoothness does not mean

The selected value is not a universal property of a time series. It depends on d , L , h , the continuation rule, the loss, and the rolling-origin design. In particular,

$$S_{d,L,h_1}^* \neq S_{d,L,h_2}^*$$

is not a contradiction; different horizons can reward different trade-offs between local adaptation and extrapolation stability.

Nor should S^* automatically be interpreted as the smoothness that best recovers a latent historical trend. Forecasting and historical signal recovery are related but distinct objectives, as predictive smoothing work such as Hart (Hart, 1994) makes clear.

7.3 | Relation to adaptive smoothing

The baseline proposal selects one smoothness value for a specified validation problem. It does not make the smoothing parameter time-varying within the fitted window. Adaptive exponential-smoothing methods such as Taylor (Taylor, 2004) answer a different question by allowing smoothing parameters to evolve with observed conditions.

A later extension could allow forecast-optimal smoothness itself to vary across origins or regimes. That requires an additional adaptation mechanism and is outside the baseline paper.

7.4 | Relation to forecast-oriented filter design

Zafar et al. (Zafar et al., 2022) show that trend-filter design can matter for short-term forecast performance. The present approach holds the filter family fixed and

optimizes its degree of smoothness. The two ideas are complementary: one changes the representation; the other changes the regularization strength inside a given representation.

7.5 | Why the smoothness scale is useful

The criterion provides a common language between smoothing and forecast validation. It lets an analyst report not only that a penalty was cross-validated, but what effective amount of smoothing the forecast task selected. It also makes horizon dependence visible on a bounded scale.

This is particularly useful when comparing series of different lengths or repeated rolling windows. Instead of comparing raw penalties, one can compare S^* and its implied effective degrees of freedom.

7.6 | Limitations

The baseline framework assumes equally spaced observations, fixed difference order, and deterministic continuation of the estimated trend. It does not model a separate stochastic residual forecast component. In applications where short-run dynamics contain substantial predictive information, a trend-only forecast is intentionally incomplete.

That limitation is useful scientifically because it isolates the effect of choosing trend smoothness for extrapolation. ARMA, state-space, or machine-learning residual models can be layered on later, but adding them to the baseline experiment would make it harder to identify whether forecast gains came from the smoothness criterion or from the residual model.

Finally, the rolling validation loss can be noisy and multimodal. The present paper treats this as part of the selection problem rather than assuming unimodality. Efficient and potentially certified recovery of multiple minima is a separate numerical contribution.

8 | CONCLUSION

Controlled-smoothness PLS makes the amount of trend regularization interpretable, but leaves open how that smoothness should be chosen when the trend is used for forecasting. This paper proposes a direct answer: choose smoothness by the chronological future-block forecast loss generated by the same penalized trend and continuation rule that will be used operationally.

The resulting object,

$$S_{d,L,h}^* \in \arg \min_{S \in [0,1]} F_{d,L,h}(S),$$

is conditional on a clearly specified forecasting configuration. The normalized smoothness scale maps the full penalty path to $[0, 1]$, has an exact effective-degrees-of-freedom interpretation, preserves the minimizers of the original penalty parameterization, and includes the two limiting smoothing models explicitly.

The proposal sits at the intersection of three established ideas: controlled smoothness in finite-difference PLS, predictive selection of smoothing and tuning parameters, and rolling out-of-sample forecast evaluation. Its contribution is their combination into a horizon-specific smoothness criterion for penalized trend forecasting.

The empirical question is therefore not whether forecast-based tuning is universally superior to classical smoothing criteria. It is when aligning the smoothness target with the actual forecast horizon changes the selected trend enough to improve later predictions. The simulation and public-data protocol in this paper is designed to answer that question while keeping recovery performance, forecast performance, and numerical optimization conceptually separate.

DATA AVAILABILITY

The methodological development does not depend on proprietary data. The submission-stage empirical evaluation is designed around public forecasting data and reproducible real-time or archived series when historical vintages matter. Any exploratory current-vintage macroeconomic results are treated separately from the submission-grade real-time-vintage analysis.

CODE AVAILABILITY

Reusable implementations of finite-difference penalized trend estimation, chronological forecast objectives, rolling-origin validation, and smoothness transformations are maintained in the `trend_estimation` package at <https://github.com/heri-espino/Trend-Estimation>. The repository also contains the manuscript source, experiment protocols, and frozen result artifacts used for the paper.

CONFLICT OF INTEREST

The author declares no conflict of interest.

References

Bergmeir, C., & Benítez, J. M. (2012). On the use of

- cross-validation for time series predictor evaluation. *Information Sciences*, 191, 192–213. doi: 10.1016/j.ins.2011.12.028
- Cortés-Toto, D., Guerrero, V. M., & Reyes, H. J. (2017). Trend smoothness achieved by penalized least squares with the smoothing parameter chosen by optimality criteria. *Communications in Statistics—Simulation and Computation*, 46(2), 1492–1507. doi: 10.1080/03610918.2015.1005236
- Franjic, D., & Schweikert, K. (2025). Predictor preselection for mixed-frequency dynamic factor models: A simulation study with an empirical application to gdp nowcasting. *Journal of Forecasting*, 44, 255–269. doi: 10.1002/for.3193
- Guerrero, V. M. (2007). Time series smoothing by penalized least squares. *Statistics & Probability Letters*, 77(12), 1225–1234. doi: 10.1016/j.spl.2007.03.006
- Guerrero, V. M. (2008). Estimating trends with percentage of smoothness chosen by the user. *International Statistical Review*, 76(2), 187–202. doi: 10.1111/j.1751-5823.2008.00047.x
- Hart, J. D. (1994). Automated kernel smoothing of dependent data by using time series cross-validation. *Journal of the Royal Statistical Society: Series B (Methodological)*, 56(3), 529–542. doi: 10.1111/j.2517-6161.1994.tb01998.x
- Staněk, F. (2023). Optimal out-of-sample forecast evaluation under stationarity. *Journal of Forecasting*, 42, 2249–2279. doi: 10.1002/for.3013
- Tashman, L. J. (2000). Out-of-sample tests of forecasting accuracy: An analysis and review. *International Journal of Forecasting*, 16(4), 437–450. doi: 10.1016/S0169-2070(00)00065-0
- Taylor, J. W. (2004). Smooth transition exponential smoothing. *Journal of Forecasting*, 23, 385–394.
- Wolff, D., & Echterling, F. (2024). Stock picking with machine learning. *Journal of Forecasting*, 43, 81–102. doi: 10.1002/for.3021
- Xu, W., Aschakulporn, P., & Zhang, J. E. (2025). Modeling and forecasting the cboe vix with the tvp-har model. *Journal of Forecasting*, 44, 1638–1657. doi: 10.1002/for.3260
- Zafar, U., Kellard, N., & Vinogradov, D. (2022). Multi-stage optimization filter for trend based short-term forecasting. *Journal of Forecasting*, 41(2), 345–360. doi: 10.1002/for.2810